

COMBINATORICS INTEGER SEQUENCES ASYMPTOTICS

Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

A proof of the OEIS A000139 parity conjecture,
with exact valuation distributions and limit laws

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Abstract

For $n \geq 0$, put

$$a_n = \frac{2(3n)!}{(n+1)!(2n+1)!}, \quad v(n) = v_2(a_n).$$

The OEIS entry A000139 records a July 2025 conjecture of Peter Bala: a_n is odd exactly when n is an odd fibbinary number. We prove the assertion in two independent ways, using binary carries and formal power series over $\mathbb{F}_2$. More generally, we determine the rational bivariate generating function of the complete valuation distribution on every interval $0 \leq n < 2^m$. It has a cubic denominator, giving a third-order recurrence in the binary length. For each fixed valuation r , the counts have an explicitly determined minimal eventual recurrence and asymptotics of order $m^{\lfloor r/2 \rfloor} \varphi^m$, with explicit constants. We also classify maximal valuations and find the least positive index of every valuation. For a uniformly chosen index below 2^m , we obtain exact mean and variance formulas, a central limit theorem, and a uniform local limit theorem. A six-state weighted digit automaton computes the full distribution below an arbitrary bound without enumerating its indices. The accompanying Python programs, exact data, and execution records reproduce the finite checks. The proofs do not depend on those checks.

Main resolution. For every $n \geq 0$,

$$a_n \text{ is odd} \iff n \text{ is odd and its binary expansion contains no } 11.$$

Status of the claim. The assertion was still labeled a conjecture in the OEIS entry inspected for this article. This is a proof of that recorded assertion, not a claim that an exhaustive literature search has established first priority. The valuation refinements are proved here as well; their priority has not been independently established.

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1 The selected problem and the scope of the result

The integer sequence

$$a_n = \frac{2(3n)!}{(n+1)!(2n+1)!} = \frac{2}{(n+1)(2n+1)} \binom{3n}{n} \quad (1)$$

begins

$$2, 1, 2, 6, 22, 91, 408, 1938, 9614, 49335, 260130, 1402440, \dots$$

For $n \geq 1$ it counts permutations sorted by two successive applications of the usual stack-sorting map. This is the deterministic stack-sorting operator, not an arbitrary two-stack machine with unrestricted choices. The same formula occurs in the enumeration of fighting fish; Fang gives a bijection between these objects and two-stack-sortable permutations [5]. The formula itself and several further interpretations are recorded in OEIS A000139 [1]. Our arithmetic arguments use only (1).

An odd *fibbinary number* is an odd nonnegative integer whose binary expansion contains no two consecutive 1s. These numbers are

$$1, 5, 9, 17, 21, 33, 37, 41, 65, 69, 73, 81, 85, \dots,$$

and form OEIS A022341 [3]. The terminology and its connection with Fibonacci representations are discussed by Lindroos, Sills, and Wang [4].

The selected question is the assertion attributed to Peter Bala, dated July 24, 2025, in A000139: the indices of the odd terms are exactly these odd fibbinary numbers [1]. The entry inspected on September 19, 2026 still used the conjecture label. Its inspected internal version identified itself as revision 297, dated May 12, 2026 [2]. That documents the provenance of the question. It does not exclude the possibility of an earlier proof elsewhere or of an unrecorded resolution.

The problem is well suited to an elementary attack because a factorial quotient converts prime divisibility into digit sums. At the prime 2, the relevant sum is $n + 2n$, so the two summands differ by just a binary shift. The apparent global arithmetic question becomes a finite-state question about adjacent digits and propagated carries. We pursue this structure rather than stopping at a parity test.

1.1 Conventions

For a positive integer b , $v_2(b)$ denotes the exponent of 2 in b . We write $s_2(n)$ for the number of 1s in the binary expansion of n , with $s_2(0) = 0$. Define

$$t(n) = v_2(n+1), \quad c(n) = \text{the total number of carries in the binary addition } n + 2n.$$

Every outgoing carry is counted, including a carry beyond the highest nonzero input digit. The number $t(n)$ is the length of the terminal run of 1s in the usual, most-significant-digit-first binary expansion.

The index $n = 0$ is retained throughout: $a_0 = 2$ and $v(0) = 1$. Thus it contributes one term of valuation 1, not one odd term. A dyadic interval always means $0 \leq n < 2^m$, and an arbitrary bound is also exclusive. Fibonacci numbers are normalized by $F_0 = 0$, $F_1 = 1$. Finally,

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \rho = \varphi^{-1}, \quad \alpha = \log_2 \varphi.$$

2 An elementary carry proof of the parity conjecture

2.1 Factorial valuations and digit sums

Lemma 2.1. *For every integer $k \geq 0$,*

$$v_2(k!) = k - s_2(k).$$

Proof. Each multiple of 2^j contributes one additional factor of 2 to $k!$, so its valuation is $\sum_{j \geq 1} \lfloor k/2^j \rfloor$. Write $k = \sum_{i \geq 0} b_i 2^i$, where $b_i \in \{0, 1\}$. Interchanging finite sums,

$$\sum_{j \geq 1} \left\lfloor \frac{k}{2^j} \right\rfloor = \sum_{i \geq 1} b_i \sum_{j=1}^i 2^{i-j} = \sum_{i \geq 0} b_i (2^i - 1) = k - s_2(k).$$

□

Proposition 2.2 (Exact valuation). *For every $n \geq 0$,*

$$v(n) = s_2(n) + s_2(n+1) - s_2(3n), \quad (2)$$

$$= 1 - t(n) + c(n). \quad (3)$$

Proof. Apply theorem 2.1 to (1). The result is

$$1 + (3n - s_2(3n)) - (n + 1 - s_2(n+1)) - (2n + 1 - s_2(2n+1)).$$

Since $s_2(2n+1) = s_2(n) + 1$, this is (2). Incrementing n changes its $t(n)$ terminal 1s into 0s and changes the next 0 into a 1. Consequently,

$$s_2(n+1) = s_2(n) + 1 - t(n). \quad (4)$$

For completeness, the carry identity needed here follows column by column. Let b_i be the digits of n , put $b_{-1} = 0$, and let q_i be the incoming carry in column i , with $q_0 = 0$. If d_i is the output digit of $3n$, then

$$b_i + b_{i-1} + q_i = d_i + 2q_{i+1}.$$

Sum this identity over all columns, including zero columns until the carry has vanished. The sums of the incoming and outgoing carries agree, and therefore

$$c(n) = \sum_{i \geq 0} q_{i+1} = 2s_2(n) - s_2(3n). \quad (5)$$

Combining (4) and (5) gives (3). □

2.2 Why the trailing run matters

It would be incorrect to infer parity solely from the absence of carries in $\binom{3n}{n}$. The factor $2/(n+1)$ in (1) changes the valuation by $1 - t(n)$. The following three cases account for this correction exactly.

Theorem 2.3 (The recorded parity conjecture). *For every $n \geq 0$, the following are equivalent:*

- (i) a_n is odd;
- (ii) n is odd and the binary expansion of n contains no adjacent 1s;
- (iii) n is odd and the bitwise expression $n \& (2n)$ is zero.

Thus the odd indices are precisely OEIS A022341.

Proof. If n is even, then $t(n) = 0$, so (3) gives $v(n) = 1 + c(n) \geq 1$.

Suppose n ends in exactly one 1, so $t(n) = 1$. Then $v(n) = c(n)$. There is no carry in $n + 2n$ exactly when the nonzero digits of the two summands do not overlap. Indeed, an overlap creates a carry, while in the absence of overlaps the incoming carry remains zero inductively. Because the second summand is a shift of the first, an overlap is exactly an adjacent pair 11 in n . Hence $v(n) = 0$ exactly in the desired case.

It remains to exclude a terminal run of length $t \geq 2$. Its digits satisfy $b_0 = \dots = b_{t-1} = 1$ and $b_t = 0$. In column 1, $b_1 + b_0 = 2$ starts a carry. That carry continues through columns $2, \dots, t-1$. In column t we still have $b_t + b_{t-1} + q_t = 0 + 1 + 1 = 2$. There are consequently at least t outgoing carries, in columns $1, \dots, t$. Equation (3) now yields $v(n) \geq 1 - t + t = 1$. This excludes every remaining odd index.

Finally, overlap of the bit supports of n and $2n$ is exactly the condition $n \& (2n) \neq 0$, proving the equivalence with (iii). $\square$

Remark 2.4 (The top carry is essential). For $n = 3$, the addition is $11_2 + 110_2 = 1001_2$. There are two carries, not one. The valuation is $1 - 2 + 2 = 1$, in agreement with $a_3 = 6$. Omitting the outgoing carry above the leading digit would give the wrong parity. This boundary condition is explicitly included in every automaton and program below.

3 An independent proof with generating functions

The ordinary generating function of the sequence can be expressed through the ternary-tree series. This representation is recorded in A000139 [2]; we derive it to make the second proof independent of an unproved generating-function identity.

Lemma 3.1. *There is a unique series $T(x) \in 1 + x\mathbb{Z}[[x]]$ such that*

$$T(x) = 1 + xT(x)^3. \quad (6)$$

Moreover, with $A(x) = \sum_{n \geq 0} a_n x^n$,

$$A(x) = T(x)(3 - T(x)). \quad (7)$$

In particular, every a_n is an integer.

Proof. Equation (6) determines each nonconstant coefficient from earlier ones, proving existence, uniqueness, and integrality recursively. Lagrange inversion, in the special form proved in section A, gives, for $k \geq 1$,

$$[x^n]T(x)^k = \frac{k}{3n+k} \binom{3n+k}{n}.$$

For $n \geq 1$, the coefficients of T and T^2 are respectively

$$\frac{(3n)!}{n!(2n+1)!}, \quad \frac{2(3n+1)!}{n!(2n+2)!}.$$

Consequently

$$[x^n](3T - T^2) = \frac{(3n)!}{n!(2n+1)!} \left(3 - \frac{3n+1}{n+1} \right) = \frac{2(3n)!}{(n+1)!(2n+1)!}.$$

The constant coefficient is $3 - 1 = 2$, as required. $\square$

Let

$$B(x) = \sum_{\substack{n \geq 0 \\ n \text{ has no binary } 11}} x^n \quad \text{in } \mathbb{Z}[[x]].$$

An admissible even integer is $2q$ for an admissible q , while an admissible odd integer is $4q + 1$ for an admissible q . Hence

$$B(x) = B(x^2) + xB(x^4). \quad (8)$$

Write bars for reduction of coefficients modulo 2. Frobenius gives $\overline{B}(x^2) = \overline{B}(x)^2$ and $\overline{B}(x^4) = \overline{B}(x)^4$. Since the constant term of B is 1, we may divide the reduced equation by $\overline{B}$ and obtain

$$1 = \overline{B} + x\overline{B}^3.$$

Thus $\overline{B}$ satisfies the same equation and initial condition as $\overline{T}$, so $\overline{B} = \overline{T}$. Reducing (7) gives

$$\overline{A} = \overline{T} + \overline{T}^2 = \overline{B} + \overline{B}^2 = x\overline{B}(x^4). \quad (9)$$

The last series has coefficient 1 exactly at the indices $4q + 1$ with q having no adjacent binary 1s. These are precisely the odd fibbinary numbers, proving theorem 2.3 a second time.

Corollary 3.2 (An algebraic parity series). *The parity series $P(x) = \overline{A}(x) \in \mathbb{F}_2[[x]]$ satisfies*

$$P(x) = x + x^2 P(x)^3. \quad (10)$$

It has algebraic degree exactly 3 over $\mathbb{F}_2(x)$.

Proof. Equation (9) says $P = x\overline{T}(x^4)$, which immediately gives (10). To show that $x^2Y^3 + Y + x$ is irreducible over $\mathbb{F}_2(x)$, it suffices to exclude a rational root. For a reduced fraction p/q with $p, q \in \mathbb{F}_2[x]$, the rational-root argument forces $p \mid x$ and $q \mid x^2$. The only candidates are $1, x, x^{-1}, x^{-2}$, none of which is a root. $\square$

4 Counting odd terms and the limits of a power-law description

Proposition 4.1. *For every $m \geq 0$,*

$$\#\{0 \leq n < 2^m : a_n \text{ is odd}\} = F_m. \quad (11)$$

Proof. For $m = 0$ the count is 0, and for $m = 1$ it is 1. When $m \geq 2$, an odd admissible binary word of length m , allowing leading zeros, ends in 01. Its remaining $m - 2$ digits can be any binary word without 11. If b_j counts such words of length j , splitting according to the initial block 0 or 10 gives $b_j = b_{j-1} + b_{j-2}$, with $b_0 = 1, b_1 = 2$. Thus $b_j = F_{j+2}$ and the desired count is F_m . $\square$

Put $O(X) = \#\{0 \leq n < X : a_n \text{ is odd}\}$ for real $X > 0$. By (11) and monotonicity between successive powers of 2,

$$O(X) = \Theta(X^\alpha), \quad \alpha = \log_2 \varphi < 1. \quad (12)$$

The odd terms therefore have natural density zero. At dyadic bounds there is the sharper exact expression

$$O(2^m) = \frac{\varphi^m - (-\rho)^m}{\sqrt{5}}.$$

One should not replace (12) by an unrestricted asymptotic $O(X) \sim CX^\alpha$ with a single positive constant. All integers in $[3 \cdot 2^{m-1}, 2^{m+1})$ begin with the binary digits 11, and so none is an odd index. Consequently

$$O(3 \cdot 2^{m-1}) = O(2^{m+1}) = F_{m+1}.$$

The two arguments have a fixed ratio $4/3$, which would contradict the ratio $(4/3)^\alpha$ predicted by such an asymptotic. The digital oscillation is a genuine feature, not an error term that disappears at all scales.

There is also a useful distinction between two kinds of generating functions. The index series

$$H(x) = \sum_{\substack{n \geq 0 \\ a_n \text{ odd}}} x^n \in \mathbb{Z}[[x]]$$

is not rational over $\mathbb{Q}$. To see this, a rational series with coefficients in the finite set $\{0, 1\}$ satisfies an eventual fixed linear recurrence; the finite vector of consecutive coefficients has only finitely many possible values and determines the next vector, so the sequence would be ultimately periodic. Here there are infinitely many 1s but also arbitrarily long zero intervals, which excludes ultimate periodicity. By contrast, the *binary-length counting series* is rational:

$$\sum_{m \geq 0} O(2^m) z^m = \frac{z}{1 - z - z^2}.$$

Confusing these two series would conceal an important change of index.

5 A finite-state model for all valuations

5.1 The three-state carry core

Read digits of n from least to most significant. A raw carry state consists of the previous input digit and the incoming carry. There are four possibilities, but $(1, 0)$ and $(0, 1)$ have identical transitions and can be merged. Let

$$A = (0, 0), \quad B = \{(1, 0), (0, 1)\}, \quad C = (1, 1).$$

Weight an outgoing carry by y . The transition table is

State	Input 0	Input 1	Terminal weight
A	A	B	1
B	A	yC	1
C	yB	yC	y

The notation yC means move to C and multiply the path weight by y . The terminal weight flushes the remaining carry through zero digits. In particular, a terminal state C creates one more carry and then moves to B , after which the carry vanishes.

The transfer matrix and terminal vector are

$$M(y) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & y \\ 0 & y & y \end{pmatrix}, \quad f(y) = \begin{pmatrix} 1 \\ 1 \\ y \end{pmatrix}. \quad (13)$$

Writing $e_A = (1, 0, 0)$, the polynomial

$$C_m(y) = e_A M(y)^m f(y) \quad (14)$$

counts all carry totals $c(n)$ for $0 \leq n < 2^m$. All length- m words are read, including those with leading zeros. Such zeros leave the final weight unchanged.

5.2 Removing the trailing-run correction without negative weights

Formula (3) contains $-t(n)$, which could suggest negative powers of y . They are unnecessary. Introduce three prefix states: S before any digit is read, U after an initial run of exactly one 1, and W while an initial run of at least two 1s is still continuing. The complete valuation automaton has six states:

State	Input 0	Input 1	Terminal weight
S	yA	U	y
U	A	W	1
W	yB	W	y
A	A	B	1
B	A	yC	1
C	yB	yC	y

Proposition 5.1. *Starting at S , the product of the transition weights and terminal weight for the binary expansion of n is exactly $y^{v(n)}$. This remains true when arbitrary leading zeros are appended to the input word.*

Proof. An even index begins with a 0. Its valuation is $1 + c(n)$, accounted for by $S \rightarrow yA$ and the carry core. An index with $t = 1$ begins with the least-significant-first block 10, leading from S through U to A with weight 1; the remaining valuation is precisely the remaining carry count. The isolated one-digit word has valuation 0, explaining the terminal weight at U .

For $t \geq 2$, the t initial 1s create $t - 1$ carries before the first zero. In $1 - t + c$, their contribution is $1 - t + (t - 1) = 0$, so the loop at W has weight 1. The terminating zero creates one additional carry and leaves the raw state $(0, 1)$, namely B . This gives the transition $W \rightarrow yB$. If the word ends before that zero is read, the mandatory flush produces exactly this same factor y , explaining the terminal weight at W .

The empty word represents $n = 0$, so its terminal weight must be y . All other terminal weights are those of the carry core. The rules also show directly that reading a further zero and then terminating has the same effect as terminating immediately, in every state. $\square$

This proposition gives a finite combinatorial interpretation with nonnegative weights for the entire valuation distribution. The extra prefix states are essential for the factorial denominator; the three-state core alone counts $c(n)$, not $v(n)$.

6 The complete dyadic distribution and its rational generating function

Define

$$V_m(y) = \sum_{0 \leq n < 2^m} y^{v(n)}, \quad \mathcal{V}(z, y) = \sum_{m \geq 0} V_m(y) z^m. \quad (15)$$

The variable z measures binary length, whereas y measures valuation.

Theorem 6.1 (Bivariate distribution). *One has*

$$\mathcal{V}(z, y) = \frac{y + (1 - y^2)z - y^3 z^2 + y(y - 1)^2 z^3}{1 - (1 + y)z - (y^2 - y + 1)z^2 + y(y + 1)z^3}. \quad (16)$$

In particular, for $m \geq 4$,

$$V_m = (1 + y)V_{m-1} + (y^2 - y + 1)V_{m-2} - y(y + 1)V_{m-3}, \quad (17)$$

with

$$V_0 = y, \quad V_1 = 1 + y, \quad V_2 = 1 + 3y, \quad V_3 = 2 + 5y + y^3. \quad (18)$$

Proof. Put

$$\begin{aligned} D(z, y) &= \det(I - zM(y)) \\ &= 1 - (1 + y)z - (y^2 - y + 1)z^2 + y(y + 1)z^3. \end{aligned} \quad (19)$$

The generating functions for carry-core paths starting at A and B are respectively

$$C(z, y) = e_A(I - zM)^{-1}f = \frac{(1 + z)(1 - yz)}{D(z, y)}, \quad (20)$$

$$R(z, y) = e_B(I - zM)^{-1}f = \frac{1 - yz + y^2z(1 - z)}{D(z, y)}. \quad (21)$$

These expressions can be verified without matrix inversion: if the three starting-state series are C, R, Q , the transition table gives

$$C = 1 + z(C + R), \quad R = 1 + z(C + yQ), \quad Q = y + zy(R + Q),$$

and solving these three linear equations gives (20) and (21).

Partition all words according to their trailing run. The empty word contributes y . The even words contribute yzC . The isolated one-bit word 1 contributes z , and the words beginning with the least-significant-first block 10 contribute z^2C . All-ones words of length at least two contribute $yz^2/(1 - z)$. Finally, a run of at least two 1s followed by its terminating zero contributes $yz^3R/(1 - z)$. Therefore

$$\mathcal{V} = y + yzC + z + z^2C + \frac{yz^2}{1 - z} + \frac{yz^3}{1 - z}R. \quad (22)$$

Substituting (20)–(21) and simplifying yields (16); the apparent factor $1 - z$ cancels. Multiplication by the denominator and extraction of z^m gives (17) for $m \geq 4$, and direct extraction gives (18). $\square$

Remark 6.2 (The starting index of the recurrence). The numerator in (16) has degree three in z . Thus the homogeneous recurrence starts at $m = 4$, not at $m = 3$. Using only three initial polynomials and starting too early loses the exceptional prefix contribution $y(y - 1)^2z^3$.

For illustration, the next polynomials are

$$\begin{aligned} V_4 &= 3 + 8y + y^2 + 3y^3 + y^4, \\ V_5 &= 5 + 13y + 2y^2 + 7y^3 + 3y^4 + 2y^5, \\ V_6 &= 8 + 21y + 4y^2 + 14y^3 + 8y^4 + 6y^5 + 3y^6. \end{aligned}$$

These show, for example, that valuation 3 appears before valuation 2. The specializations

$$\mathcal{V}(z, 0) = \frac{z}{1 - z - z^2}, \quad \mathcal{V}(z, 1) = \frac{1}{1 - 2z} \quad (23)$$

recover the Fibonacci odd count and the total count 2^m .

7 Every fixed valuation: exact recurrences and asymptotics

For a fixed integer $r \geq 0$, set

$$b_r(m) = [y^r]V_m(y), \quad H_r(z) = \sum_{m \geq 0} b_r(m)z^m.$$

Write

$$d(z) = 1 - z - z^2. \quad (24)$$

The denominator in (16) has the particularly useful form

$$D(z, y) = (1 - yz)d(z) - y^2z^2(1 - z). \quad (25)$$

This explains why the order of the fixed-level recurrence increases only after every second valuation.

7.1 A finite closed expression for each level

Let

$$N_0 = z, \quad N_1 = 1 + z^3, \quad N_2 = -z - 2z^3, \quad N_3 = z^3 - z^2,$$

so that the numerator of (16) is $\sum_{j=0}^3 N_j y^j$. For $s \geq 0$, define

$$J_s(z) = z^s \sum_{j=0}^{\lfloor s/2 \rfloor} \binom{s-j}{j} \frac{(1-z)^j}{d(z)^{j+1}}, \quad J_s = 0 \quad (s < 0). \quad (26)$$

Then

$$H_r(z) = \sum_{j=0}^3 N_j(z) J_{r-j}(z). \quad (27)$$

Indeed, expand (25) as a geometric series:

$$\frac{1}{D} = \frac{1}{d} \sum_{j \geq 0} \frac{y^{2j} z^{2j} (1-z)^j}{d^j (1-yz)^{j+1}}.$$

The coefficient of y^s is exactly (26), proving (27). This is a nonrecursive finite formula for every fixed-level generating function.

The first four simplify to

$$H_0(z) = \frac{z}{d}, \quad (28)$$

$$H_1(z) = \frac{1 + z^2 + z^3}{d}, \quad (29)$$

$$H_2(z) = \frac{z^4(1 - z^2)}{d^2}, \quad (30)$$

$$H_3(z) = \frac{z^3(1 - z^2)(1 + z + z^2)}{d^2}. \quad (31)$$

For $r \geq 4$ they also satisfy the simple recurrence in valuation level

$$H_r = zH_{r-1} + \frac{z^2(1-z)}{d}H_{r-2}. \quad (32)$$

All these formulas are identities of formal series; no convergence assumption is being made.

Theorem 7.1 (Minimal eventual recurrences). *Let $k = \lfloor r/2 \rfloor$. In lowest terms over $\mathbb{Q}(z)$, the denominator of $H_r(z)$ is exactly*

$$(1 - z - z^2)^{k+1}. \quad (33)$$

Hence $b_r(m)$ has a minimal eventual constant-coefficient recurrence of order $2k + 2$, with characteristic polynomial

$$(X^2 - X - 1)^{k+1}.$$

Proof. Formula (27) shows that the denominator divides d^{k+1} . To prove that this power cannot be lowered, it is enough to show a pole of order $k + 1$ at the positive root ρ of d . For $r = 2k$, only the term $N_0 J_{2k}$ contributes to that highest order; its numerator at ρ is

$$\rho^{2k+1}(1 - \rho)^k = \rho^{4k+1} > 0.$$

For $r = 2k + 1$, the two contributing terms are $N_0 J_{2k+1}$ and $N_1 J_{2k}$. Their combined numerator at ρ is

$$\begin{aligned} (k+1)\rho^{2k+2}(1 - \rho)^k + (1 + \rho^3)\rho^{2k}(1 - \rho)^k \\ = \rho^{4k-1}(1 + k\rho^3) > 0. \end{aligned}$$

Here $1 - \rho = \rho^2$ and $1 + \rho = \rho^{-1}$ were used. Since d is irreducible over $\mathbb{Q}$, this proves (33). A rational generating function in lowest terms has precisely that denominator as its minimal eventual recurrence polynomial, with the coefficients reversed. $\square$

7.2 Explicit leading asymptotics and full expansions

Theorem 7.2 (Fixed-level asymptotics). *For every fixed $k \geq 0$, as $m \rightarrow \infty$,*

$$b_{2k}(m) \sim \frac{\rho^{3k}}{5^{(k+1)/2} k!} m^k \varphi^m, \quad (34)$$

$$b_{2k+1}(m) \sim \frac{\rho^{3k-2}(1 + k\rho^3)}{5^{(k+1)/2} k!} m^k \varphi^m. \quad (35)$$

More precisely, for each r there are polynomials P_r, Q_r of degree at most $\lfloor r/2 \rfloor$ such that, for all sufficiently large m ,

$$b_r(m) = P_r(m)\varphi^m + Q_r(m)(-\rho)^m. \quad (36)$$

Their coefficients are obtained explicitly by partial fractions of (27).

Proof. Factor

$$d(z) = (1 - z/\rho)(1 + \rho z).$$

At $z = \rho$, the second factor is $1 + \rho^2 = \sqrt{5}\rho$. The highest-pole numerators computed in theorem 7.1 therefore give

$$\begin{aligned} H_{2k}(z) &\sim \frac{\rho^{3k}}{5^{(k+1)/2}} (1 - z/\rho)^{-k-1}, \\ H_{2k+1}(z) &\sim \frac{\rho^{3k-2}(1 + k\rho^3)}{5^{(k+1)/2}} (1 - z/\rho)^{-k-1}. \end{aligned}$$

The coefficient of $(1 - z/\rho)^{-k-1}$ is $\binom{m+k}{k} \rho^{-m} \sim m^k \varphi^m / k!$, proving the displayed asymptotics. The only other pole is at $-\varphi$. Partial fractions at the two poles, together with a possible polynomial part of finite support, give (36). This also provides all lower powers of m in the asymptotic expansion. $\square$

For $k \geq 1$, the error after retaining just the leading term is $O(m^{k-1}\varphi^m)$. For $k = 0$ there are exact Fibonacci descriptions: $b_0(m) = F_m$ for all m , and $b_1(m) = F_{m+2}$ for $m \geq 2$. The exceptions in the latter statement are $b_1(0) = b_1(1) = 1$.

Corollary 7.3 (Almost all terms have any prescribed fixed power of 2). *For each fixed $R \geq 1$, the set of indices n such that $2^R \mid a_n$ has natural density 1. More quantitatively, with $k = \lfloor (R-1)/2 \rfloor$,*

$$\#\{0 \leq n < X : 2^R \nmid a_n\} = \Theta((\log X)^k X^\alpha).$$

Proof. At $X = 2^m$, sum the finitely many levels $0 \leq r < R$ and use theorem 7.2. The largest polynomial degree is k , and every leading constant is positive. Monotonicity between adjacent powers of 2 gives the arbitrary-bound estimate. Since $\alpha < 1$, its ratio to X tends to zero. $\square$

These are fixed- r results. They are not uniform estimates when r grows proportionally to m . The central regime requires a different analysis, given in section 9.

8 Maximal valuations and the least index of every valuation

Theorem 8.1 (Sharp dyadic maximum). *For every $m \geq 3$,*

$$\max_{0 \leq n < 2^m} v(n) = m, \quad \#\{0 \leq n < 2^m : v(n) = m\} = F_{m-2}. \quad (37)$$

Equality $v(n) = m$ occurs exactly for the even indices $n = 2q$ such that q has binary length $m-1$, ends in 11, and contains no 00. The leading digit of q is, as usual, 1.

Proof. If an integer q has binary length ℓ , its addition to $2q$ has no carry in column 0 and has at most one outgoing carry in each column $1, \dots, \ell$. Thus $c(q) \leq \ell$.

For even $n = 2q$, shifting does not change the number of carries, so $v(n) = 1 + c(q) \leq m$. To have equality, q must have length $\ell = m-1$ and a carry in every column $1, \dots, \ell$. The first such carry requires its two lowest digits to be 11. Once the carry is present, it survives a column exactly when the two adjacent input digits are not both 0. The leading digit 1 ensures the final outgoing carry. This proves the stated characterization for even indices.

If n is odd with $t(n) \geq 2$, then $v(n) = 1 - t(n) + c(n) \leq m-1$. If $t(n) = 1$, the two lowest digits are 01. No carry can occur in columns 0, 1, 2; equivalently, after deleting the least-significant-first block 10, the remaining carry computation starts at A . Hence $v(n) \leq m-2$ for $m \geq 2$. No odd index is an extremizer.

It remains to count the described words q . Delete their last 1. The resulting words have length $m-2$, start and end in 1, and contain no 00. The number of length- j words with these properties is F_j : for $j = 1, 2$ it is 1, and splitting after the initial 1 gives the Fibonacci recurrence. Therefore there are F_{m-2} extremizers, and there is at least one for each $m \geq 3$. $\square$

For $m = 0, 1, 2$, the maximum is 1, as the initial polynomials show. The transition from these exceptional cases to (37) is why the least occurrence of valuation 2 is later than that of valuation 3.

Theorem 8.2 (Least positive indices). *Let ℓ_r be the least positive integer n such that $v(n) = r$. Then*

$$\ell_0 = 1, \quad \ell_1 = 2, \quad \ell_2 = 13, \quad (38)$$

and for all $r \geq 3$,

$$\ell_r = \frac{2^{r+1} + 6 + 4(-1)^r}{3}. \quad (39)$$

Proof. The valuations at $n = 1, \dots, 13$ are

$$0, 1, 1, 1, 0, 3, 1, 1, 0, 1, 3, 3, 2,$$

which proves (38). For $r \geq 3$, theorem 8.1 shows that no index of fewer than r binary digits can have valuation r . Among indices of length r , the extremal-word characterization applies. The smallest eligible binary word is obtained lexicographically by choosing 0 whenever it does not create 00 or violate the final 11 condition on q . Consequently the binary word of ℓ_r is

$$\begin{cases} (10)^{(r-3)/2}110, & r \text{ odd,} \\ (10)^{(r-4)/2}1110, & r \text{ even.} \end{cases}$$

Summing the corresponding geometric progressions gives $(2^{r+1} + 2)/3$ in the odd case and $(2^{r+1} + 10)/3$ in the even case, which is (39). $\square$

The beginning of this least-index sequence is

$$1, 2, 13, 6, 14, 22, 46, 86, 174, 342, 686, 1366, \dots$$

Its tail is a linear combination of 2^r , 1, and $(-1)^r$. Accordingly, for $r \geq 6$,

$$\ell_r = 2\ell_{r-1} + \ell_{r-2} - 2\ell_{r-3}.$$

Here “least” refers to a positive index. Allowing the index 0 would replace $\ell_1 = 2$ by 0 and leave the other values unchanged.

9 Exact moments and Gaussian limit laws

Let N_m be uniformly distributed on $\{0, 1, \dots, 2^m - 1\}$ and put $X_m = v(N_m)$. Its probability generating function is

$$\mathbb{E}[y^{X_m}] = 2^{-m} V_m(y). \quad (40)$$

This is a random *index* model. It is not a model that first chooses a permutation uniformly among all objects of several different sizes.

9.1 Exact mean and variance

Theorem 9.1. For all $m \geq 0$,

$$\mathbb{E}X_m = \frac{m}{2} - \frac{2}{3} + \frac{9 + (-1)^m}{6 \cdot 2^m}. \quad (41)$$

For $m \geq 1$,

$$\text{Var } X_m = \frac{11m}{12} - \frac{23}{9} + \frac{27 - (-1)^m}{6 \cdot 2^m} - \frac{(9 + (-1)^m)^2}{36 \cdot 4^m}. \quad (42)$$

For $m = 0$, $X_0 = 1$ deterministically and its variance is 0.

Proof. Differentiate (16) with respect to y and set $y = 1$. The resulting first-moment series has the partial-fraction form

$$\sum_{m \geq 0} V'_m(1)z^m = -\frac{7}{6(1-2z)} + \frac{1}{2(1-2z)^2} + \frac{1}{6(1+z)} + \frac{3}{2(1-z)}. \quad (43)$$

Thus

$$V'_m(1) = 2^m \left(\frac{m}{2} - \frac{2}{3} \right) + \frac{3}{2} + \frac{(-1)^m}{6},$$

which proves (41).

Differentiating twice gives the factorial second moment. The full partial-fraction identity is recorded in section B; its coefficient is

$$\begin{aligned} V''_m(1) = 2^m \left(\frac{m^2}{4} - \frac{m}{4} - \frac{13}{9} \right) + (-1)^m \left(\frac{m}{6} - \frac{5}{9} \right) \\ + \frac{3m}{2} + 1 + \mathbf{1}_{m=0}. \end{aligned} \quad (44)$$

Since $\mathbb{E}X_m^2 = 2^{-m}(V''_m(1) + V'_m(1))$, subtracting the square of (41) and simplifying gives (42) for $m \geq 1$. The indicator in (44) accounts for the exceptional constant coefficient when $m = 0$. $\square$

In particular,

$$\mathbb{E}X_m = \frac{m}{2} - \frac{2}{3} + O(2^{-m}), \quad \text{Var } X_m = \frac{11m}{12} - \frac{23}{9} + O(2^{-m}). \quad (45)$$

The variance is substantially larger than $m/4$, the variance of a sum of m independent fair binary digits. Carry propagation creates correlations; merely counting independent local patterns would miss this effect.

9.2 The dominant eigenvalue and its symmetry

The characteristic polynomial of (13) is

$$\chi(\lambda, y) = \lambda^3 - (1+y)\lambda^2 - (y^2 - y + 1)\lambda + y(y+1). \quad (46)$$

At $y = 1$ it factors as $(\lambda - 2)(\lambda - 1)(\lambda + 1)$. Let $\lambda(y)$ be the analytic root with $\lambda(1) = 2$. For real $y > 0$ near 1, it is the Perron eigenvalue of the nonnegative, irreducible, aperiodic matrix $M(y)$.

The simple pole corresponding to this root in (16) gives analytic functions $h(y)$ and an $\eta < 2$, on a sufficiently small complex neighborhood of 1, such that

$$V_m(y) = h(y)\lambda(y)^m + O(\eta^m), \quad h(1) = 1. \quad (47)$$

The error is uniform in a smaller closed neighborhood. This follows either from the spectral projections of $M(y)$ or directly by partial fractions in z , since the three roots remain separated there. The polynomial part in z affects only finitely many initial m .

Let J be the permutation matrix swapping states A and C . A direct calculation gives

$$JM(y)J = yM(1/y).$$

Perron eigenvalues for positive real y , followed by analytic continuation near 1, therefore satisfy

$$\lambda(y) = y\lambda(1/y). \quad (48)$$

Put

$$K(s) = \log \frac{\lambda(e^s)}{2}.$$

Equation (48) says that $K(s) - s/2$ is even. In particular, all its odd Taylor coefficients beyond the first vanish. Implicit differentiation of (46) gives

$$\lambda'(1) = 1, \quad \lambda''(1) = \frac{4}{3},$$

and hence

$$K'(0) = \frac{1}{2}, \quad K''(0) = \frac{11}{12}. \quad (49)$$

A further algebraic expansion, reproducible by the symbolic verification script, gives

$$K(s) = \frac{s}{2} + \frac{11s^2}{24} - \frac{569s^4}{1728} + O(s^6). \quad (50)$$

For example, one can first solve (46) in powers of $u = y - 1$:

$$\lambda(1+u) = 2 + u + \frac{2}{3}u^2 - \frac{1}{3}u^3 - \frac{7}{27}u^4 + O(u^5),$$

and then substitute $u = e^s - 1$. Symmetry supplies the vanishing fifth coefficient. The usual analytic generating-function framework for such limit laws is described in [6]; the needed argument is included next rather than invoked as a black box.

9.3 Central and local limit theorems

Theorem 9.2. *As $m \rightarrow \infty$,*

$$\frac{X_m - m/2}{\sqrt{11m/12}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (51)$$

The following stronger local statement holds uniformly for integers r :

$$\sup_{r \in \mathbb{Z}} \left| \sqrt{\frac{11m}{12}} \mathbb{P}(X_m = r) - \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(r - m/2)^2}{2(11m/12)}\right) \right| \rightarrow 0. \quad (52)$$

One may replace $m/2$ by the exact mean in either statement.

Proof. Write $\sigma^2 = 11/12$ and $\mu = 1/2$. For fixed real t , substitute $y = \exp(it/(\sigma\sqrt{m}))$ into (47). After multiplying by $\exp(-it\mu\sqrt{m}/\sigma)$, (49) yields the limit $\exp(-t^2/2)$ for the characteristic function of the normalized variable. The prefactor $h(y)$ tends to 1, and the error is exponentially small. The continuity theorem for characteristic functions proves (51).

We verify the extra spectral condition for a local theorem. For $|u| = 1$, entrywise absolute values give $|M(u)| = M(1)$, so every eigenvalue has modulus at most 2. Suppose $M(u)w = \gamma w$ with $|\gamma| = 2$. Taking absolute values gives $2|w| \leq M(1)|w|$. Multiplication by the positive left Perron vector forces equality in every coordinate. By irreducibility, $|w|$ is a positive Perron vector, and equality also holds in every triangle inequality used above. The self-loop at A , whose weight is 1, then forces $\gamma/2 = 1$. The self-loop at C , whose weight is u , forces $u = 1$. Therefore

$$\text{spr}(M(e^{it})) < 2 \quad \text{for } 0 < |t| \leq \pi. \quad (53)$$

On a closed arc bounded away from 0, compactness makes this a uniform strict gap. The recurrence (17), whose characteristic roots are those of M , consequently bounds $V_m(e^{it})/2^m$ uniformly by $C\theta^m$ there, for

some $\theta < 1$. A resolvent contour with radius strictly between the maximum root modulus and 2 gives this bound even at a repeated subdominant eigenvalue.

Near $t = 0$, (49) and analyticity give constants $c, \delta > 0$ such that

$$\left| \frac{\lambda(e^{it})}{2} \right|^m \leq e^{-cmt^2} \quad (|t| \leq \delta).$$

Fourier inversion gives

$$\mathbb{P}(X_m = r) = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2^{-m} V_m(e^{it}) e^{-irt} dt.$$

Set $s = \sigma\sqrt{m}t$ and remove the phase corresponding to μm . On each bounded s -interval, (47) gives convergence of the integrand to $e^{-s^2/2}$. The preceding Gaussian bound dominates the part with $|t| \leq \delta$, and (53) makes the remaining part negligible even after multiplication by $\sqrt{m}$. Thus the rescaled characteristic functions converge in L^1 to the Gaussian characteristic function, with the finite integration interval extended by zero. Fourier inversion and the bound of the difference of transforms by the L^1 difference prove (52), uniformly in r .

Finally, (41) differs from μm by a bounded amount. Dividing that difference by $\sigma\sqrt{m}$ gives zero in the limit, and the Gaussian density is uniformly continuous. This justifies either choice of center. $\square$

The local theorem describes the central distribution on the scale $\sqrt{m}$. It does not provide relative errors in the extreme tails. In particular, the rare fixed-valuation regime is governed by theorem 7.2, not by substituting a fixed r into a Gaussian approximation and expecting relative accuracy.

10 Algorithms, independent checks, and reproducibility

10.1 Evaluating a single index

The exact valuation requires no construction of a_n :

```
def valuation(n: int) -> int:
    if not isinstance(n, int) or isinstance(n, bool) or n < 0:
        raise ValueError("n must be a nonnegative integer")
    return n.bit_count() + (n + 1).bit_count() - (3*n).bit_count()
```

The parity criterion is the still simpler predicate

```
(n & 1) != 0 and (n & (n << 1)) == 0
```

These operate on the digits of n , whereas a_n itself has a number of digits proportional to n . The six-state automaton supplies an alternative digit-by-digit implementation, independent of the population-count expression.

10.2 The complete distribution below an arbitrary bound

The recurrence (17) computes all $V_0, \dots, V_m$ with $O(m^2)$ integer coefficient updates. An arbitrary interval $[0, N)$ can also be handled without enumeration. Process candidate digits and the digits of N from least to most significant, maintaining the six-state automaton and a comparison state in $\{-1, 0, 1\}$.

After the lowest j digits have been read, the comparison state is the comparison of those two low-digit integers. At the next position, an equal pair of bits leaves the comparison unchanged; an unequal pair *overrides* it, because that new bit is more significant than every previously read bit. After all positions, retain only comparison -1 . This gives an 18-state digit dynamic program. Attach a polynomial in y to each state and apply the weights from section 5. The terminal weights complete the valuation.

For $L = \text{bits}(N)$ this uses $O(L^2)$ integer coefficient updates and stores $O(L)$ integer coefficients across a constant number of states. These are arithmetic-operation counts, not unit-cost bit-complexity claims: the coefficients themselves may have $O(L)$ bits. The implementation checks that the final sum of histogram entries equals N exactly. The case $N = 0$ is handled separately by the zero histogram.

10.3 What was actually executed

The archive contains a standard-library implementation `code/carries.py`, a standard-library verification driver `code/verify.py`, and an optional symbolic verifier using SymPy. The full exact test run recorded the following checks:

Check	Executed range
Direct factorial quotient against digit valuation, carry count, automaton valuation, and parity	$0 \leq n \leq 2000$
Digit valuation and parity; histograms against the recurrence	$0 \leq n < 2^{20}$
Independent column-carry count and six-state automaton against digit sums	$0 \leq n < 2^{18}$
Every arbitrary-bound digit-DP histogram against enumeration	$0 \leq N \leq 2048$
Digit DP against dyadic recurrence; exact moments, parity counts, and extremal coefficients	$0 \leq m \leq 256$
Least positive index, by exhaustive search	$0 \leq r \leq 18$
Formula witnesses of valuation r , without exhaustive minimality testing	$3 \leq r \leq 1000$
Non-dyadic large-bound digit DP and exact total	$N = 10^{100}$

All these checks passed. The executed environment used Python 3.13.5; the full standard-library verification run took about 3.3 seconds in this session. Timing is environment-specific and is not a performance guarantee. The exact report, including the measured duration, is in `data/verification.json`.

These comparisons deliberately distinguish levels of independence. The million-index test does not compute a million factorial quotients; it compares the proved digit formula with the binary predicate and the independently implemented distribution recurrence. Direct quotient checks stop at 2000. Likewise, evaluating the formula for ℓ_{1000} checks that its valuation is 1000, but does not exhaust all smaller integers. Minimality at every level rests on theorem 8.2.

The symbolic verifier checks the rational identities, fixed-level formulas, moment decompositions, and the eigenvalue expansion. It is a useful algebraic cross-check, not a machine-checked proof of the Perron–Frobenius or Fourier arguments. No Lean formalization or external peer review is claimed.

10.4 Reproduction commands

From the extracted archive directory, run

```
python code/verify.py  
python code/carries.py --n 13  
python code/carries.py --dyadic 100  
python code/carries.py --bound 100000000000000000000000000000000  
python code/carries.py --least 1000  
python code/verify_symbolic.py  
pdflatex -interaction=nonstopmode -halt-on-error research.tex  
pdflatex -interaction=nonstopmode -halt-on-error research.tex
```

The symbolic script is optional and requires SymPy. The main program and exact verification driver require no third-party packages and make no network calls. Generated CSV files contain the first 201 sequence terms, every dyadic histogram through $m = 256$, and least indices through valuation 64. The large-bound JSON contains the complete valuation histogram for $0 \leq n < 10^{100}$.

11 Conclusion, provenance, and remaining directions

The selected parity assertion is established by [theorem 2.3](#), and [section 3](#) gives an independent proof in formal power series. Its mathematical explanation is especially short: factorial valuations are digit sums, the sum $n + 2n$ is controlled by adjacent digits, and the factor $n + 1$ forces careful treatment of the trailing run of 1s.

The same analysis gives substantially more than parity. The full dyadic valuation distribution has the rational generating function (16). The cubic denominator governs a recurrence in binary length, fixed valuation levels have exact denominators $d^{\lfloor r/2 \rfloor + 1}$, and the dominant poles give their asymptotic expansions. A different dominant-eigenvalue calculation gives the exact moment constants and the Gaussian central regime. Extremal words give both the Fibonacci number of maximizers and the explicit least-index formula. The algorithms directly implement these finite-state proofs.

The status statement should remain precise. The OEIS page provides a recent, explicitly conjectural target; the article proves that target. Targeted searches for A000139 parity and valuation, fibbinary indices, and parity of two-stack-sortable permutations did not locate a prior resolution in the material retrieved. This is not an exhaustive bibliographic certification. The classical enumeration formula, the ternary-tree generating function, and the fibbinary terminology are not claimed as new. Nor is the general method of using finite matrices to study digital statistics new. The specific identities proved here are presented with explicit derivations so that their correctness can be assessed independently of any priority claim.

Several natural extensions remain outside the scope of the proofs. One is a parity-reversing involution directly on two-stack-sortable permutations whose fixed objects explain the fibbinary indices. Another is the joint distribution of valuations at different primes, where the binary shift structure no longer supplies this particular three-state core. A third is a uniform saddle-point analysis when r/m stays away from both 0 and the central value $1/2$, connecting the fixed-level and central regimes through explicit large-deviation asymptotics. Finally, the full normalized counting oscillation for non-dyadic bounds can be studied through the digit dynamic program rather than reduced to upper and lower bounds. No solution of these additional questions is asserted here.

A The special Lagrange inversion identity used above

We give the formal calculation to keep theorem 3.1 self-contained. Let $U = x\Phi(U)$ with $\Phi(0) = 1$. For $n \geq 1$ and a formal power series G , the coefficient formula is

$$[x^n]G(U(x)) = \frac{1}{n}[u^{n-1}]G'(u)\Phi(u)^n. \quad (54)$$

Indeed, using formal residues and substituting $x = u/\Phi(u)$ gives

$$\begin{aligned} [x^n]G(U(x)) &= \text{Res}_{u=0} G(u) \left(\frac{\Phi(u)}{u} \right)^{n+1} \frac{\Phi(u) - u\Phi'(u)}{\Phi(u)^2} du \\ &= \text{Res}_{u=0} G(u) \left(\frac{\Phi(u)^n}{u^{n+1}} - \frac{\Phi(u)^{n-1}\Phi'(u)}{u^n} \right) du. \end{aligned}$$

The expression in parentheses is $-\frac{1}{n}\frac{d}{du}(\Phi(u)^n/u^n)$. Formal integration by parts, whose validity is the fact that a formal derivative has zero residue, proves (54). The residue substitution is valid because $u/\Phi(u) = u + O(u^2)$ has a compositional inverse; checking monomials gives the usual change-of-variable rule.

For $T = 1 + U$ and $\Phi(u) = (1 + u)^3$, take $G(u) = (1 + u)^k$. Then

$$[x^n]T^k = \frac{k}{n} \binom{3n+k-1}{n-1} = \frac{k}{3n+k} \binom{3n+k}{n}.$$

For $n = 0$ both sides are 1, so the same expression also holds there.

B Second-moment algebra and a compact symbolic certificate

The second derivative of (16) at $y = 1$ is

$$\begin{aligned} \sum_{m \geq 0} V_m''(1)z^m &= 1 - \frac{17}{18(1-2z)} - \frac{1}{(1-2z)^2} + \frac{1}{2(1-2z)^3} \\ &\quad - \frac{13}{18(1+z)} + \frac{1}{6(1+z)^2} - \frac{1}{2(1-z)} + \frac{3}{2(1-z)^2}. \end{aligned} \quad (55)$$

Extracting coefficients using

$$[z^m](1-az)^{-j} = \binom{m+j-1}{j-1}a^m$$

gives (44). This also makes the isolated constant 1 on the right, and hence the $m = 0$ correction, explicit.

A compact exact certificate for the principal generating-function identity is the polynomial equality

$$D(z, y) \left(y + yzC + z + z^2C + \frac{yz^2}{1-z} + \frac{yz^3R}{1-z} \right) = y + (1-y^2)z - y^3z^2 + y(y-1)^2z^3,$$

with C, R as in (20)–(21). Multiplication by $1 - z$ turns it into a literal identity in $\mathbb{Z}[z, y]$. The included symbolic script checks this identity after exact cancellation, along with the fixed-level formulas and moment identities. Thus the displayed rational function is not inferred by fitting a finite initial segment.

C A small exact table

The following table records $b_r(m)$ for the first few binary lengths. Blank entries are zero. Each row sums to 2^m ; its first entry is F_m .

$m \backslash r$	0	1	2	3	4	5	6	7	8
0	0	1							
1	1	1							
2	1	3							
3	2	5	0	1					
4	3	8	1	3	1				
5	5	13	2	7	3	2			
6	8	21	4	14	8	6	3		
7	13	34	8	27	16	16	9	5	
8	21	55	15	51	32	34	25	15	8

The first valuations at individual indices are

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$v(n)$	1	0	1	1	1	0	3	1	1	0	1	3	3	2	4	1

These tables are for orientation, not substitutes for the all-index arguments. The CSV artifacts provide larger exact tables.

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